

Molecular dynamics simulation of cyclic plasticity in irradiated FeCoNiCrCu high entropy alloy

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ABSTRACT

High-entropy alloys (HEAs) have attracted significant attention due to their exceptional mechanical properties and remarkable radiation resistance. However, structural materials in reactors are subjected not only to irradiation damage but also to fatigue loads induced by thermal cycling and vibration during their service life. In this study, we investigate the mechanical response and microstructural evolution of the FeCoNiCrCu HEA under cyclic loading after cascade overlapping simulations, using molecular dynamics simulations. The irradiation process was simulated at various primary knock-on atom energies up to 20 keV to introduce defect structures. Our simulations reveal the specific atomistic mechanisms driving the cyclic hardening behavior in HEAs: lattice disordered atoms play a dual role, acting as localized stress concentrators for dislocation nucleation while simultaneously serving as strong pinning sites that contribute to the initial yield strength. As cyclic deformation progresses, the evolution of the dislocation network governs the global transition between softening and hardening. In the later stages of cycling, a surge in stacking fault fraction is observed, which is identified as the dominant mechanism driving the sustained cyclic hardening. These findings provide atomic-scale insights into the correlation between irradiation defects and fatigue performance in HEAs. While the high strain rate may bias dislocation and stacking-fault accumulation quantitatively, the qualitative mechanisms identified remain robust.

1. Introduction

High-entropy alloys (HEAs) have attracted significant attention as potential structural materials for advanced nuclear systems due to their exceptional radiation resistance, high strength, and ductility [1–6]. While experimental quantitative studies confirm these advantages, real-time observation of atomic-scale defect dynamics remains challenging. Consequently, researchers have increasingly relied on molecular dynamics simulations to uncover the mechanisms governing radiation tolerance. Previous simulations indicate that the unique lattice distortion and complex composition of HEAs enhance defect recombination and retard damage accumulation [7–13]. He et al. [14] systematically demonstrated that irradiation resistance improves with increasing compositional complexity. Specifically, the FeCoNiCrCu HEA exhibits the highest defect recombination rate among the studied systems, attributed to the extended duration of the thermal spike phase induced by the high-entropy effect. Han et al. [15] highlighted the

sluggish diffusion effect in FeCoNiCrCu, demonstrating that the reduced mobility of dislocation loops hinders their interaction and growth, thereby resulting in a finer defect structure compared to conventional metals. However, reactor components must also endure cyclic mechanical loads during service [16]. Previous work by Luzginova et al. [17] on irradiated steels highlighted a crossover effect, where irradiation improves fatigue life at low strain amplitudes due to hardening, but reduces it at high amplitudes due to ductility loss. Sato et al. [18] utilized positron annihilation lifetime spectroscopy to characterize defect structures in Ni. They highlighted that while unirradiated fatigue mainly produces single vacancies, irradiated fatigue involves vacancy clusters originating from cascades, whose growth is notably hindered by deformation-induced dislocation networks. Although the interaction between irradiation defects and fatigue has been studied in traditional materials, the behavior of HEAs under such coupled conditions remains underexplored.

Based on this context, this paper employs molecular dynamics

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simulation methods to investigate the mechanical response and microstructural evolution mechanisms of FeCoNiCrCu HEAs under cyclic loading conditions, subsequent to overlapping cascade simulations. To achieve this, we performed cascade simulations to introduce irradiation defects, followed by cyclic loading simulations to observe the resultant mechanical and structural changes. The study particularly aims to elucidate the interaction mechanisms between irradiation defects and cyclic deformation, attempting to reveal these interrelations at an atomic scale, thereby providing insights into the fatigue performance evolution of HEAs post-irradiation and supplying mechanistic support for their application in advanced nuclear systems.

2. Methodology

2.1. Scheme for cascade simulation

Molecular dynamics (MD) simulations of irradiation cascades were conducted utilizing the Large-scale Atomic/Molecular Massively Parallel Simulator (LAMMPS) [19]. The interatomic interactions were modeled using the latest embedded atom method which is an improvement upon the FeCoNiCrCu HEAs potential proposed by Farkas and Caro [20]. By incorporating a smooth interpolation with the Ziegler-Biersack-Littmark [21] potential in the short-range, the potential enables accurate predictions of defect energy and displacement behavior in both pure metals and high-entropy alloys during cascade collisions. The validity of this potential had been sufficiently demonstrated in the research by Deluigi et al. [7], with the total energy E_i of an atom i :

$$E_i = F_\alpha \left(\sum_{j \neq i} \rho_j(r_{ij}) \right) + \frac{1}{2} \sum_{j \neq i} \phi_{\alpha\beta}(r_{ij}) \quad (1)$$

The energy E_i of atom i is comprised of two components: the embedding energy F_α , which denotes the energy of embedding the atom within the electron density environment provided by surrounding atoms, and the pair potential $\phi_{\alpha\beta}$, representing the direct interaction energy between atom i and its neighboring atom j .

The open-source visualization tool (OVITO) [22] was utilized to analyze the nucleation and evolution of defects, while the atomic local crystal structures were characterized using the Common Neighbor Analysis [23] method. The Dislocation Extraction Algorithm (DXA) [24] was employed to identify dislocation lengths and types, and the Wigner-Seitz defect analysis [25] was used to identify vacancies and interstitial atoms.

In terms of modeling, a single-crystal pure Ni simulation box with dimensions of $178 \text{ \AA} \times 178 \text{ \AA} \times 178 \text{ \AA}$ (corresponding to a $50 \times 50 \times 50$ unit cell, encompassing a total of 500,000 atoms) was first constructed,

with its lattice orientation aligned with the [100], [010], and [001] directions. Subsequently, the ATOMSK tool [26] was used to create an equimolar FeCoNiCrCu alloy by randomly substituting Ni atoms in the Ni lattice, and the initial lattice constant was set at $3.56 \text{ \AA}$. Following energy minimization via the steepest descent algorithm, the system was equilibrated at 300 K for 20 ps using a Nosé-Hoover thermostat.

In the displacement cascade simulations, a stratified region division strategy was implemented to balance computational efficiency and physical fidelity. The system was divided into: (i) an outer fixed boundary layer to absorb shockwaves; (ii) an intermediate thermostat region (NVT) to dissipate heat; and (iii) a central active region (NVE) for the cascade dynamics, as illustrated in Fig. 1(a). To mitigate boundary effect interference, before each simulation, the entire simulation box was randomly translated along the x , y and z directions, and a primary knock-on atom (PKA) was randomly selected in the central region. To avoid channeling effects, the PKA was given initial kinetic energy in the high-index [135] direction as indicated in Fig. 1(b). The conversion of atomic energy to velocity is governed by the formula:

$$E = \frac{1}{2} mv^2 \quad (2)$$

Where E is the energy, m is the atomic mass, and v is the atomic velocity of the incident atom.

To investigate the material response, 50 cascade simulations were performed at 300 K with PKA energies of 1, 5, 10, and 20 keV. While reactor environments involve elevated temperatures, the 300 K condition was chosen to isolate the material response by preserving the damage structure and decoupling thermal softening, thereby establishing a rigorous athermal baseline. Regarding energy selection, although neutron spectra reach the MeV range, the 20 keV upper limit is justified by the subcascade splitting phenomenon [27]. Since high-energy impacts decompose into separated lower-energy sub-cascades, this range effectively captures the representative primary damage states while ensuring the cascade volume remains contained within the simulation boundaries. The detailed simulation process is illustrated in Fig. 2. In this setup, the number of atomic displacements induced by each cascade collision was estimated using the NRT model [28] with the formula:

$$N_{NRT} = \frac{0.8E_{PKA}}{2E_d} \quad (3)$$

Consequently, the cumulative damage dose in displacements per atom (dpa) is calculated as:

$$dpa = \frac{k \times N_{NRT}}{N_{total}} \quad (4)$$

Where N_{NRT} represents the theoretical number of Frenkel pairs

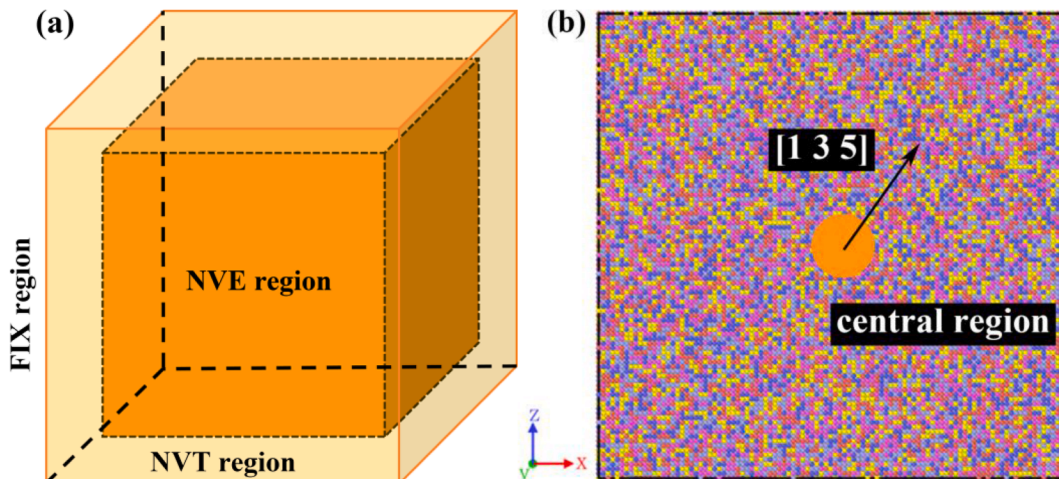


Fig. 1. (a) Diagram of simulation region division, (b) Single crystal FeCoNiCrCu HEA.

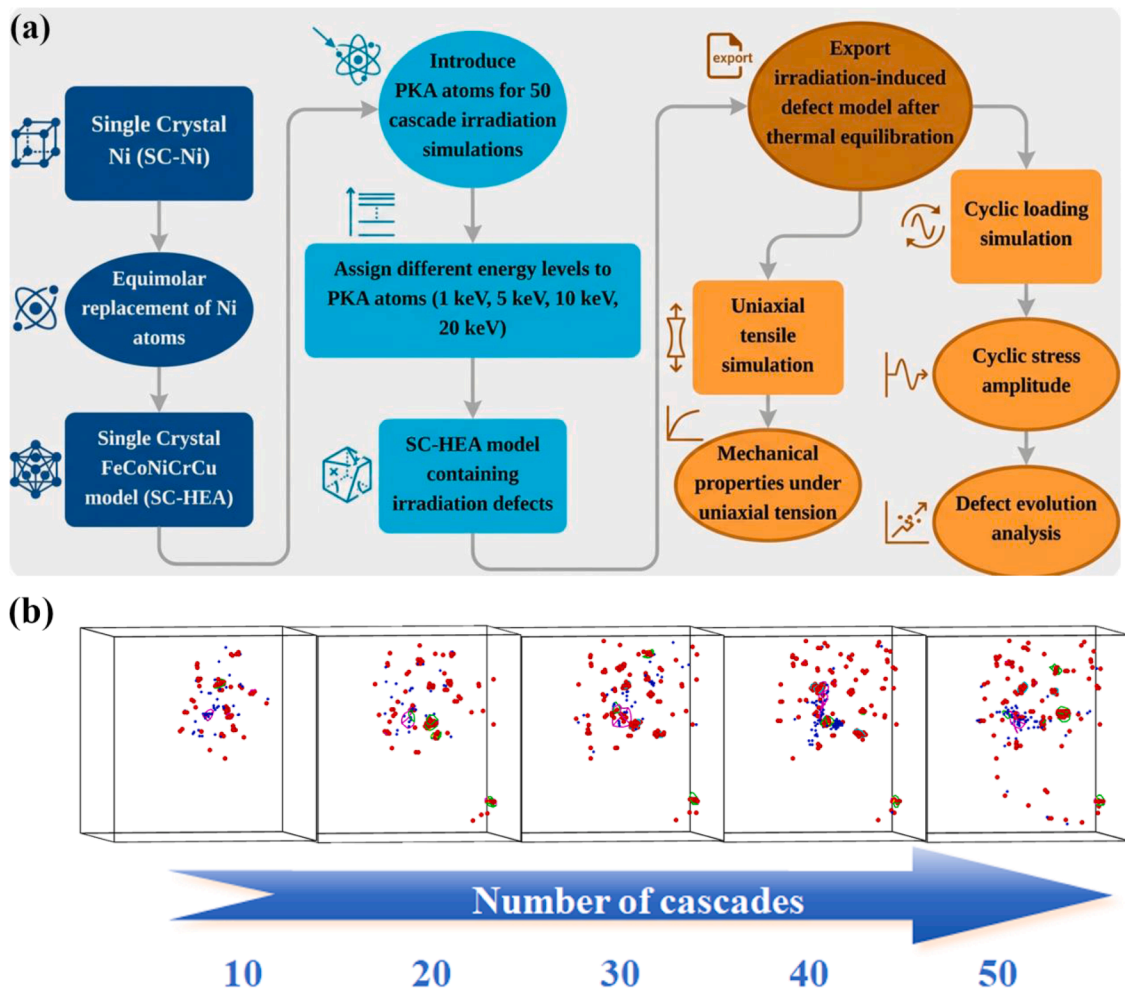


Fig. 2. (a) Simulation flowchart, (b) Diagram of overlapping cascade irradiation (red atoms are interstitials, blue atoms are vacancies).

generated per cascade, E_{PKA} is the kinetic energy of the primary knock-on atom, and E_d is the average displacement threshold energy. Furthermore, k denotes the number of accumulated cascade events, and N_{total} is the total number of atoms in the simulation system. For the FeCoNiCrCu HEA, an average displacement threshold energy (E_d) of 45.0 eV was adopted. This choice is strongly supported by recent molecular dynamics calculations for FeCoNiCrCu HEA, which determined an average E_d of 45 ± 2 eV based on extensive sampling of primary knock-on atoms in random directions [7]. Although E_d may exhibit temperature dependence [29] and the present simulations were performed at 300 K, the value of 45.0 eV is used here as a conventional reference parameter in the NRT formalism to define a nominal damage dose for comparison purposes. By construction, the NRT-dpa is an estimate of displacement production rather than a direct measure of the number of surviving defects. Therefore, the dpa values reported here should be interpreted as nominal dose metrics for cascade accumulation and comparison with previous studies, rather than as an exact prediction of the stable defect population under the present temperature conditions. This choice is also consistent with a previous study on the same FeCoNiCrCu alloy at 300 K [15]. Accordingly, the nominal dpa values corresponding to the present irradiation conditions are 0, 0.00088, 0.0044, 0.0088, and 0.0176, respectively.

Interstitial and vacancy atoms are common point defects in materials [30]. To establish irradiated HEA-FeCoNiCrCu models with varying levels of damage, the changes in the number of Wigner-Seitz defects under different irradiation doses were statistically analyzed [31].

As shown in the Fig. 3, a single cascade collision at different PKA

energies primarily undergoes four stages. Initially, a small number of defects are generated, followed by a thermal spike phase where the defect count reaches its peak, then a collision remnant phase, and finally a stage where defects recombine and stabilize [32]. To validate the reliability of the irradiation simulations, the number of surviving Frenkel pairs (N_{FP}) was benchmarked against the Norgett-Robinson-Torrens (NRT) theoretical prediction (N_{NRT}). Specifically, for a PKA energy of 10 keV, the calculated defect survival efficiency ($\eta = N_{FP}/N_{NRT}$) is 0.29. This value falls well within the reported range of 0.2–0.4 for similar FCC-HEAs [33]. Although MD simulations inherently yield higher instantaneous defect concentrations than experimental equilibrium states due to nanosecond-scale relaxation limits, this quantitative agreement confirms that the current EAM potential and PKA parameters reasonably capture the primary radiation damage mechanisms prior to the subsequent cyclic loading.

Fig. 4 illustrates the variation in defect number with the number of cascades for HEA-FeCoNiCrCu under different dosages. For low-energy PKAs (1–10 keV), the number of Frenkel pairs increases gradually and tends to saturate after approximately 30 cascades, reaching around 150 pairs. However, when the PKA energy reaches 20 keV, this saturation state is disrupted. The observed phenomena can be explained by the fact that under low-energy irradiation, the severe lattice distortion in HEAs obstructs point defect migration through high-energy barrier diffusion paths, thereby promoting recombination [34,35].

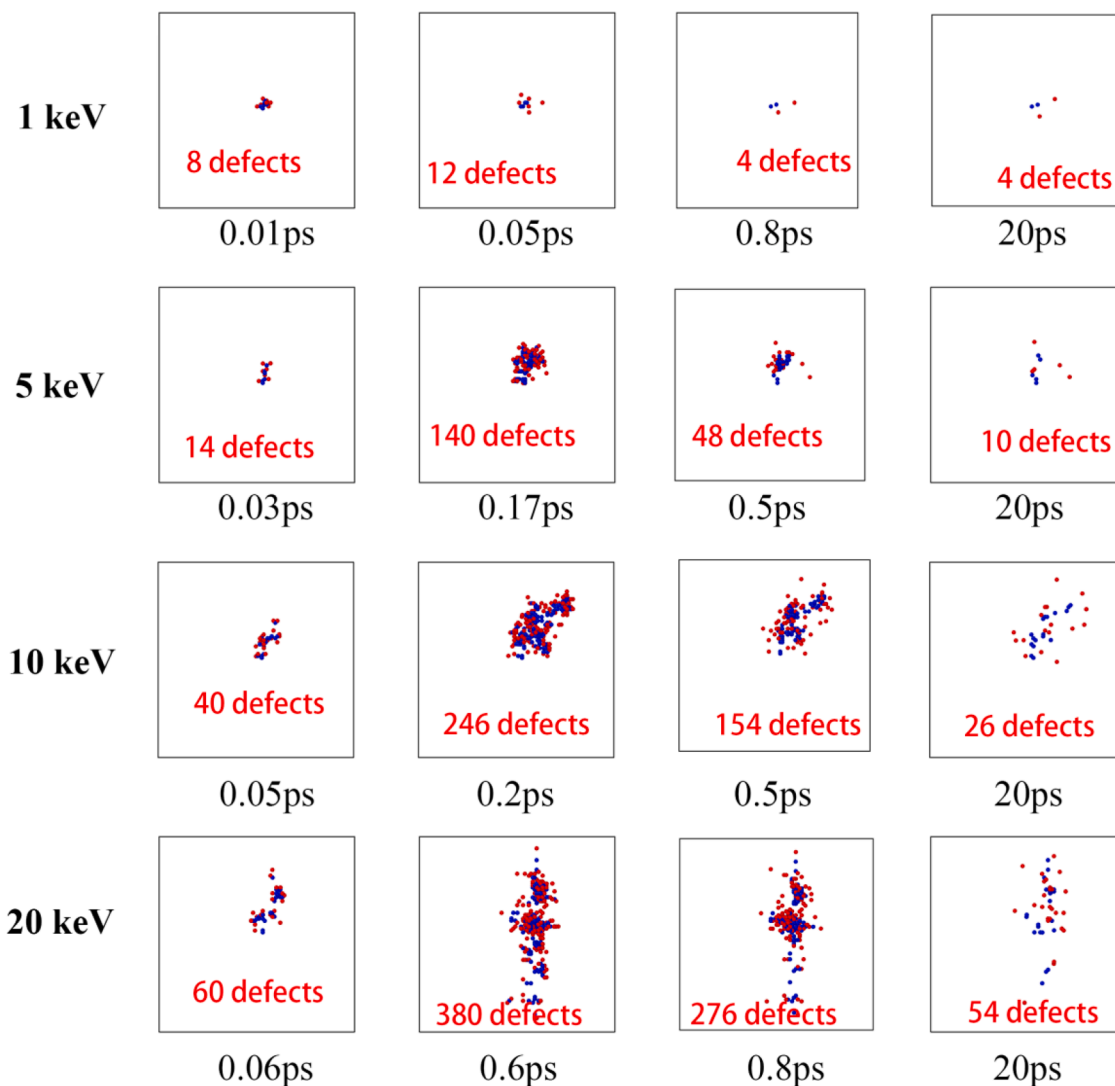


Fig. 3. Evolution of the number of defects during a single cascade collision at different PKA energies.

2.2. Cyclic deformation simulation protocol

The irradiated samples obtained at 300 K were subjected to uniaxial tensile tension and cyclic loading in the z-direction using MD simulations. The time integration step was set to 1 fs. Despite low macroscopic strain rates in cyclic plasticity, crystalline deformation remains intrinsically spatiotemporally localized [36]. In particular, Stinville et al. [37] demonstrated that fatigue-relevant slip is highly localized and occurs via discrete intermittent (burst-like) events within nanoscale active volumes. Consistently, nanoscale deformation proceeds via discrete, high-velocity dislocation avalanches [38,39], inducing local strain amplitudes that vastly exceed macroscopic averages [40]. Notably, cyclic loading is physically distinct from monotonic tension because load reversal promotes dislocation reversibility and rearrangement, rather than purely unidirectional dislocation accumulation. Nevertheless, dislocation motion remains one of the primary carriers of plastic deformation in both loading modes. Therefore, MD provides a suitable tool to investigate the spatiotemporal localization phenomena underlying macroscopic cyclic plastic deformation. It should be noted, however, that this spatiotemporal-localization rationale pertains primarily to individual elementary slip events, whereas cyclic deformation additionally involves longer-timescale processes, including dislocation reciprocation, annihilation, and progressive cell-structure formation. As a result, some quantitative metrics may carry rate-dependent bias.

In MD simulations, typical strain rates range from 10^7 to 10^{10} s^{-1} [41]. For instance, Shen et al. [42] investigated the fatigue damage formation mechanisms of single-crystal and polycrystalline aluminum at strain rates of 5×10^7 and $5 \times 10^9 \text{ s}^{-1}$, revealing that the yield stress, yield strain, and elastic modulus remained within the same order of magnitude. Based on these findings and the cyclic-loading results reported by Chen et al. [43], which showed that increasing the strain rate quantitatively raises the dislocation density and stacking-fault fraction without altering the primary deformation mechanisms, they subsequently adopted $5 \times 10^9 \text{ s}^{-1}$ for their fatigue loading simulations. Furthermore, this same study [43] also explicitly noted that MD simulations commonly operate in the range of 10^7 – 10^{10} s^{-1} , citing related atomistic studies using strain rates up to 10^{10} s^{-1} . Although a strain rate of $1 \times 10^{10} \text{ s}^{-1}$ exceeds the maximum rate ($2 \times 10^9 \text{ s}^{-1}$) directly examined in their work, it remains within the typical range used in MD simulations and can therefore be regarded as a reasonable extrapolation. In addition, Wang et al. [44] evaluated FeCoNiCrCu under various strain rates and observed minimal variations in yield stress, supporting the feasibility of using a strain rate of $1 \times 10^{10} \text{ s}^{-1}$ in MD simulations. Therefore, the rate of $1 \times 10^{10} \text{ s}^{-1}$ was adopted in our subsequent uniaxial tension and cyclic loading simulations to qualitatively elucidate the post-irradiation deformation mechanisms of FeCoNiCrCu, acknowledging that absolute quantitative values may exhibit rate-dependent deviations.

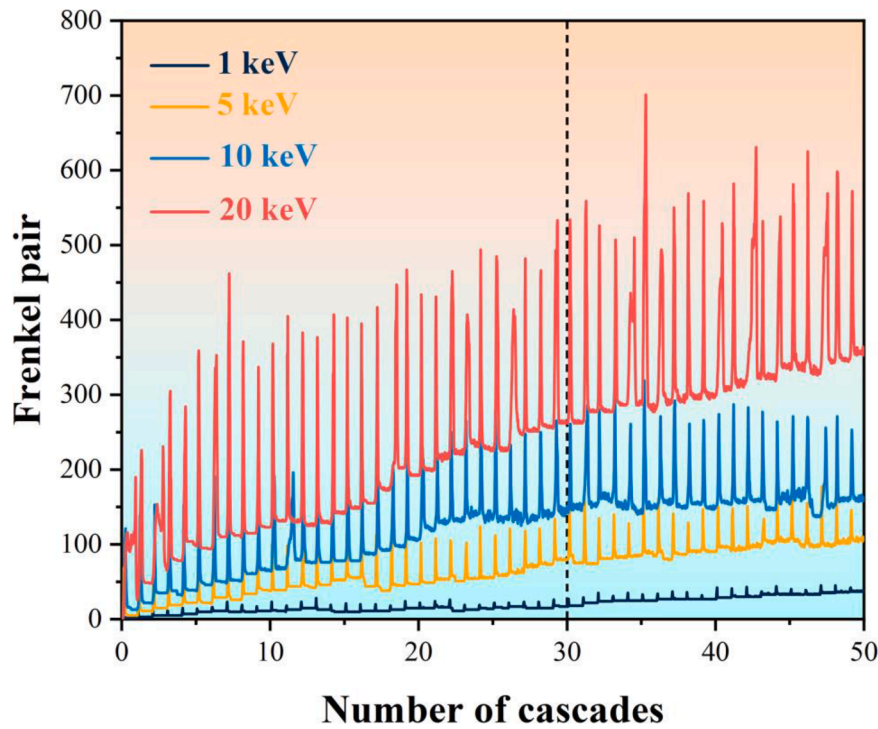


Fig. 4. Curve of Frenkel defect pairs changing with cascade count under different PKA energies.

Cyclic loading was applied to the irradiated samples, following the loading mode shown in Fig. 5. The strain amplitude was set to 0.1 (10 %) with a strain ratio of -1 . This amplitude was selected based on the stress-strain response from our uniaxial tensile simulations, which shows a monotonic yield strain of $\sim 17.8\%$. Choosing an amplitude below the monotonic yield level allows fatigue-induced cyclic accumulation to be resolved while avoiding overly rapid failure at higher amplitudes; in contrast, lower amplitudes tend to produce much weaker

irreversible evolution within the same simulation time. Therefore, 20 tension-compression cycles were performed to resolve the early-stage cyclic response and its associated microstructural evolution within a computationally feasible window.

In the initialization phase of the loading, energy minimization of the irradiated damaged model was first performed using the conjugate gradient algorithm to eliminate residual interatomic stresses. This was followed by isothermal relaxation at 300 K for 50 ps to ensure that the

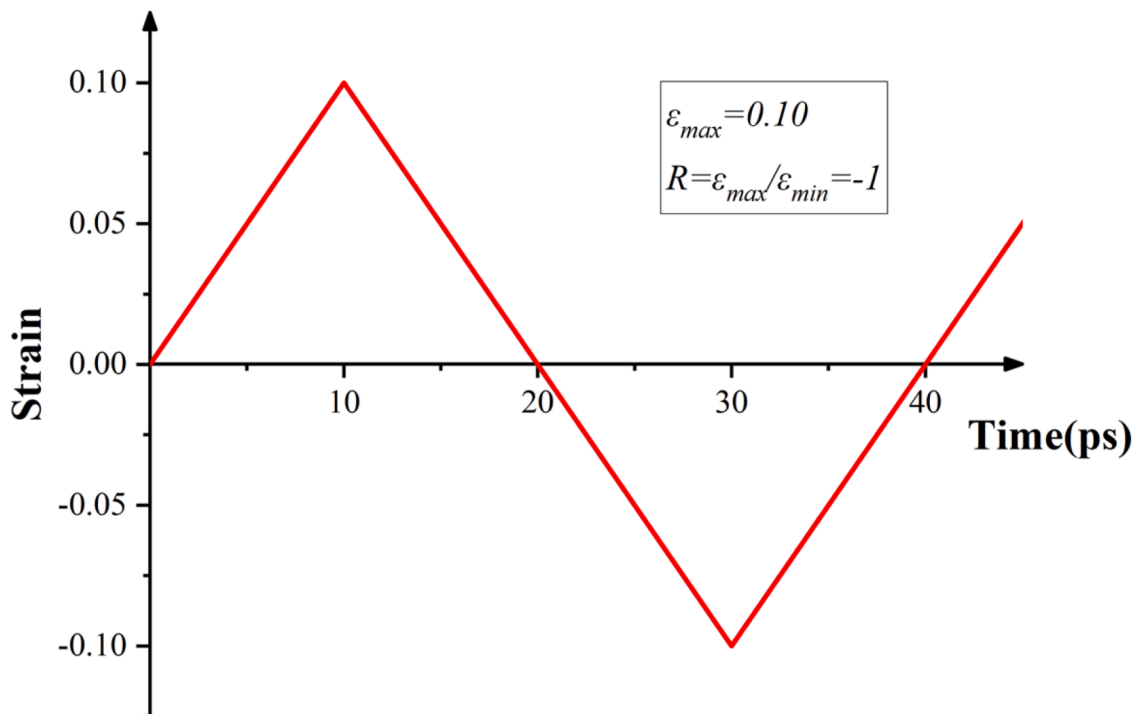


Fig. 5. Diagram of cyclic loading application.

system reached a thermodynamic equilibrium state. Strain control was achieved by dynamically adjusting the z-direction lattice parameters of the simulation box. Cyclic loading followed a sinusoidal triangular waveform loading path. Each cycle included symmetric tensile-compressive deformation: the first 10 ps applied positive strain up to +0.1, followed by 20 ps of reverse loading down to -0.1.

3. Results and discussions

3.1. Mechanical properties

3.1.1. Tensile stress-strain curve

When uniaxial tensile stress is applied to a material, irradiation-induced defects and dislocations within the material interact in complex ways, significantly altering the mechanical properties of the materials [45]. Uniaxial tensile simulations were first performed on FeCoNiCrCu HEA models with varying levels of initial irradiation damage to explore the material's mechanical properties.

The stress-strain curves under uniaxial tension for irradiated and unirradiated FeCoNiCrCu HEA at room temperature (300 K) are shown in Fig. 6(a). The deformation process can be distinctly categorized into three stages. Stage I, the elastic stage is characterized by a linear stress-strain relationship obeying Hooke's law, where the lattice deforms reversibly. Stage II, a distinct yielding stage marks the transition from elastic to plastic behavior, accompanied by a deviation from linearity as dislocation sources begin to activate. Stage III, the material enters the plastic flow stage, where stress continues to evolve nonlinearly with strain due to the complex interplay between dislocation multiplication

and motion. The stress-strain curves for all four simulated samples had identical slopes in this phase, indicating that cascade damage had not significantly altered the Young's modulus of the FeCoNiCrCu alloy. Young's modulus is primarily influenced by factors such as material composition, temperature, and microstructure. Liu [46] studied the temperature effect and initial plastic deformation of the mechanical properties of FeCoNiCrCu HEA under uniaxial tensile load, finding that Young's modulus and tensile strength increase with decreasing temperature. In this study, all materials had the same chemical composition and were simulated in the same environment, with microstructural differences being the primary influencing factor. As shown in Fig. 6(b), the distribution of the microphases remains remarkably stable with increasing irradiation dose, leading to no significant change in Young's modulus. This high phase stability is consistent with the experimental findings of Kiran et al. [47], who performed grazing incidence XRD on Fe-Ni-Mn-Cr HEAs. They reported that the alloy retained its single-phase FCC structure without any detectable second phase or amorphization,

Table 1

Mechanical properties of unirradiated FeCoNiCrCu and irradiated FeCoNiCrCu subjected to different PKA energies.

Material	ϵ_{yield}	$\sigma_{\text{yield}}(\text{GPa})$	E(GPa)
unirradiated	17.8 %	19.1	107.9GPa
irradiated-1keV	14.3 %	14.3	105.6GPa
irradiated-5keV	13.6 %	13.2	104.8GPa
irradiated-10keV	13.3 %	13.0	104.5GPa
irradiated-20keV	12.9 %	12.4	104.2GPa

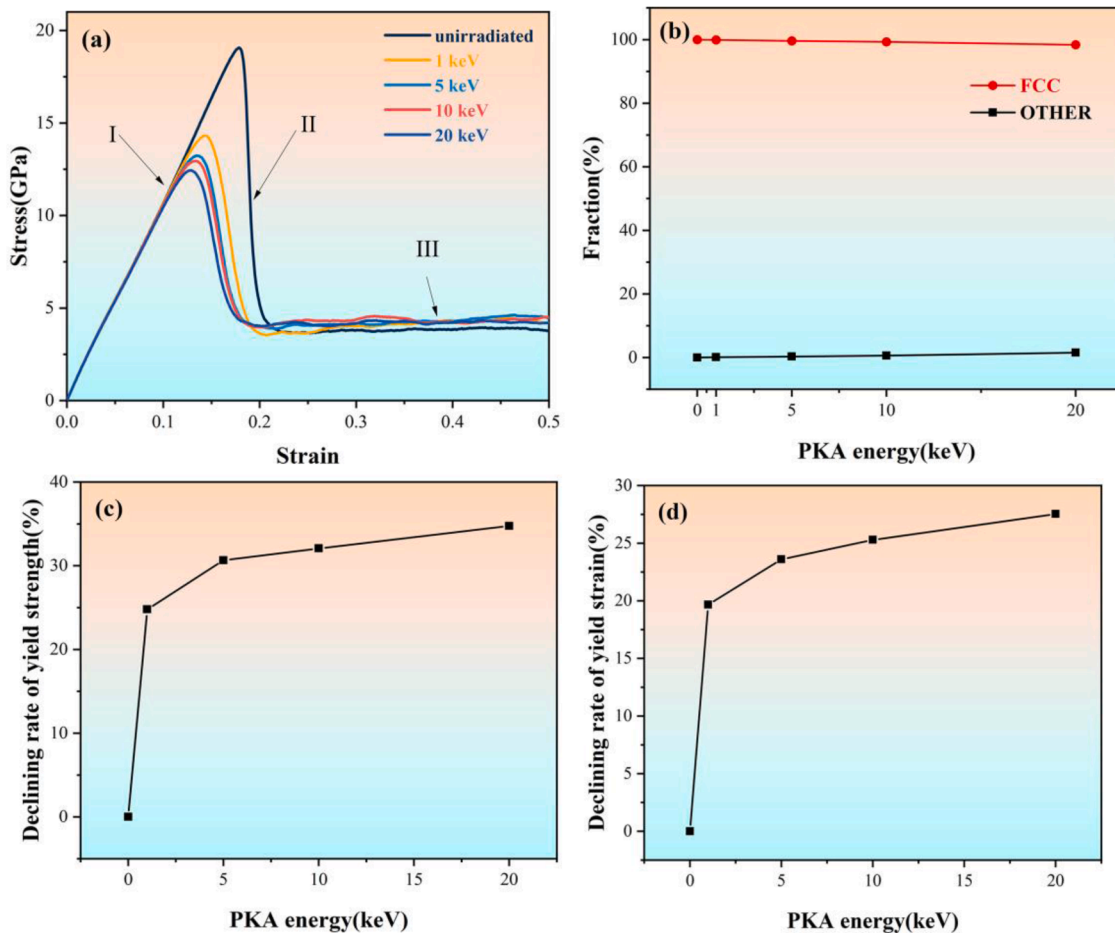


Fig. 6. (a) Uniaxial stress-strain curve, (b) Variation of face-centered cubic (FCC) phase proportion with PKA energy, (c) Reduction rate of yield strength, (d) Reduction rate of yield strain.

even under heavy ion irradiation up to 3 dpa at various temperatures.

Subsequently, as shown in Table 1, both the yield strength and yield strain of the FeCoNiCrCu HEA decreased monotonically with increasing PKA energy. Notably, the 20 keV sample yielded earliest (12.45 GPa, 0.129), indicating that irradiation leads to a premature onset of plastic deformation and diminishes the material's ductility capability. These results are consistent with previous studies [48,49]. Specifically, Liang et al. [50] revealed that this degradation is primarily attributed to cascade-induced point defects acting as nucleation sites for dislocations under applied strain. This mechanism induces premature plastic deformation and alters the deformation mode, ultimately resulting in a significant deterioration of mechanical properties. Consequently, the presence of irradiation damage significantly shortens the elastic phase, leading to an earlier onset of plastic yielding. The reduction rates of yield strength and yield strain are illustrated in Fig. 6(c) and (d). It should be noted that the lines connecting data points serve as visual guides only and do not imply a continuous mathematical relationship. The rates of decrease in yield strength and yield strain followed a similar trend; both dropped sharply with increasing irradiation dose, with the rate of decline gradually leveling off at higher doses.

The results suggest that the mechanical performance of FeCoNiCrCu changed relatively little after irradiation, with this characteristic being more pronounced at high irradiation dose.

To verify the reliability of the simulation settings regarding the mechanical response, Table 2 compares the mechanical parameters of unirradiated FeCoNiCrCu HEA obtained from uniaxial tension in this study with reported data from MD simulations. The results indicate that the yield strength and strain from our MD simulations align well with other simulation studies in terms of both trend and order of magnitude, despite inherent variations due to different interatomic potentials, exact compositions, and strain rates. It is noteworthy that the absolute yield strength values from MD simulations are consistently two orders of magnitude higher than experimental measurements for nominally similar alloys [51–53]. This significant discrepancy is an expected and well-documented consequence of the extremely high strain rates (10^8 – 10^{10} s⁻¹) required in MD.

3.1.2. Cyclic stressresponse

Plastic deformation induced by cyclic loading is crucial for assessing cyclic damage [57]. The cyclic stress-strain curves under cyclic loading for FeCoNiCrCu HEA with different irradiation doses are shown in Fig. 7 (a)–(e). Regardless of irradiation, the area of the stress-strain hysteresis loops for all materials increased with the number of cycles. The increase in the area of the stress-strain loops directly resulted from the increased plastic deformation component with each cycle. This suggested that irreversible microstructural evolution and damage accumulation occurred within the material[58].

Compared to the unirradiated samples, the irradiated materials exhibited larger areas enclosed by the stress-strain loops. The area enclosed by the stress-strain loops served as an indicator of the material's energy dissipation capacity under cyclic loading. This indicated that irradiated FeCoNiCrCu dissipated more energy during cyclic loading, likely due to more obstacles within the irradiated material, making dislocation movement more difficult.

It was also evident that the unirradiated FeCoNiCrCu HEA primarily underwent elastic deformation during the first five cycles. During this phase, the material exhibited low energy consumption and high

reversibility in its microstructure, with no significant permanent deformation. After the fifth cycle, the area enclosed by the stress-strain curves increased, indicating a transition in deformation behavior from primarily elastic to plastic deformation. Compared to traditional fatigue experiment data, the transition from elastic to plastic deformation occurred after only five cycles, which was fewer than in real experimental scenarios. This was primarily due to the inherent time scale limitations of molecular dynamics simulations [59]. For irradiated FeCoNiCrCu HEA, even fewer cycles were required for this deformation behavior transition.

The evolution of stress amplitude at the constant strain amplitude can reflect the fatigue mechanical characteristics of the material. Cyclic loading-induced softening and hardening are critical aspects of alloy cyclic plastic deformation behavior [60].

It could be observed that under cyclic loading, all materials initially underwent varying degrees of softening, followed by alternating cyclic softening and hardening. Fig. 8 illustrates the relationship between stress amplitude and average stress amplitude with the number of cycles for each material, where the black curve represents the stress amplitude for each cycle, and the red curve denotes the average stress amplitude. For ease of observation regarding the stress amplitude change as materials stably enter the plastic phase, the stress amplitudes for all materials were statistically analyzed between 7 and 20 cycles. With increasing irradiation dose, FeCoNiCrCu displayed a trend of hardening influenced by irradiation: the average stress amplitude for unirradiated material was 5.45 GPa, with stress amplitude fluctuations around the average. When PKA energy increased to 1 keV, the average stress amplitude slightly decreased to 5.41 GPa, similar to the unirradiated state, though the trend changed (in the early cycles, stress amplitudes were generally below the average, stabilizing above the average in later cycles). Moreover, with the increase in PKA energy, the average stress amplitude also increased accordingly, as shown in Fig. 8(f). Notably, all irradiated samples exhibited a pronounced increase in stress amplitude between cycles 15 and 18, which may be attributed to a stronger impediment to dislocation glide during this stage.

Monitoring of the stress-strain response confirmed that the unirradiated samples reached a cyclic saturation state after approximately 10 cycles, characterized by stabilized stress amplitudes. In contrast, for the irradiated samples, a continuous cyclic hardening behavior was observed throughout the 20 cycles. This phenomenon is attributed to the persistent interaction between dislocations and irradiation-induced defects, which impedes dislocation motion and delays the onset of saturation.

Consequently, the increase in average stress amplitude with PKA energy indicates an irradiation-induced hardening effect, which becomes more pronounced at higher energy levels.

In low-cycle fatigue under constant strain, cyclic hardening and softening are associated with dislocation movement within the crystal. Dislocation interactions and entanglements impede movement, manifested as cyclic hardening under cyclic loading. When dislocation obstacles are disrupted or dislocations undergo annihilation and reorganization, cyclic softening is observed under cyclic loading. The evolution of dislocations during cyclic loading will be discussed in the next section.

Table 2

Comparison of mechanical properties of unirradiated FeCoNiCrCu HEA between this study and reported MD simulations.

Material	Strain rate (s ⁻¹)	Temperature (K)	ϵ_{yield} (%)	σ_{yield} (GPa)	Method	Reference
This study	1×10^{10}	300K	17.8	19.1	MD Simulation	-
FeCoNiCrCu	1×10^{10}	300K	18.4	18.9	MD Simulation	Cao Y et al. [54]
FeCoNiCrCu	5×10^8	300K	16.7	18.3	MD Simulation	Liu X et al. [55]
FeCoNiCrCu	1×10^{10}	300K	16.6	22.6	MD Simulation	Wang X et al. [56]

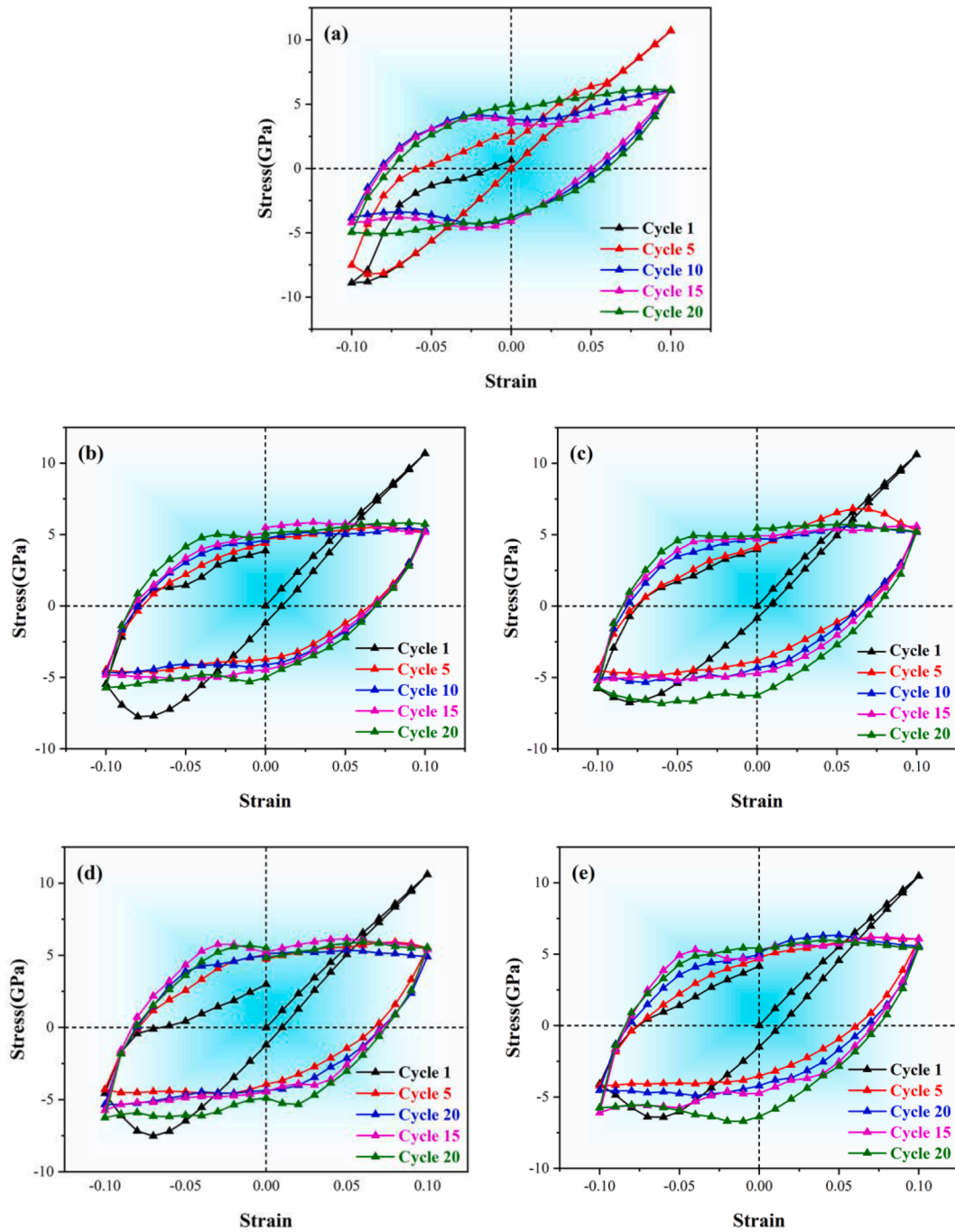


Fig. 7. Cyclic stress-strain loops for unirradiated FeCoNiCrCu and irradiated FeCoNiCrCu subjected to different PKA energies of 1k eV (b), 5k eV (c), 1kKeV (d), and 20k eV (e).

3.2. Irradiation damage evolution

3.2.1. Irradiation dislocations

The irradiation process introduces a substantial number of dislocations within the crystal, and low-cycle fatigue is fundamentally a plastic deformation process controlled by the formation and motion of these dislocations. Thus, the evolution of dislocation structures directly reflects the evolution of the material's microstructure.

Fig. 9 shows snapshots of the defect structures in FeCoNiCrCu HEA at different irradiation doses by DXA, where an increase in the total dislocation line length within the material was evident with increased irradiation dose. The evolution behavior of these initial dislocation structures under subsequent cyclic loading was the fundamental reason

for the cyclic hardening or softening of the material.

Dislocation nucleation is generally considered the initiation of plastic deformation. Analysis of the dislocation character reveals that the irradiation-induced dislocations are primarily mediated by Shockley partial dislocations. Furthermore, owing to the dissociation of perfect dislocations and reactions between Shockley partials, other sessile dislocations were also identified, including stair-rod dislocations (purple segments), Hirth dislocations (yellow segments), and Frank loops (single circular loops marked in light blue). Notably, in the HEA system, a limited number of stair-rod dislocations were generated, with the majority arranging into tetrahedral structures. These specific configurations have been widely corroborated by transmission electron microscopy (TEM) observations [61–63].

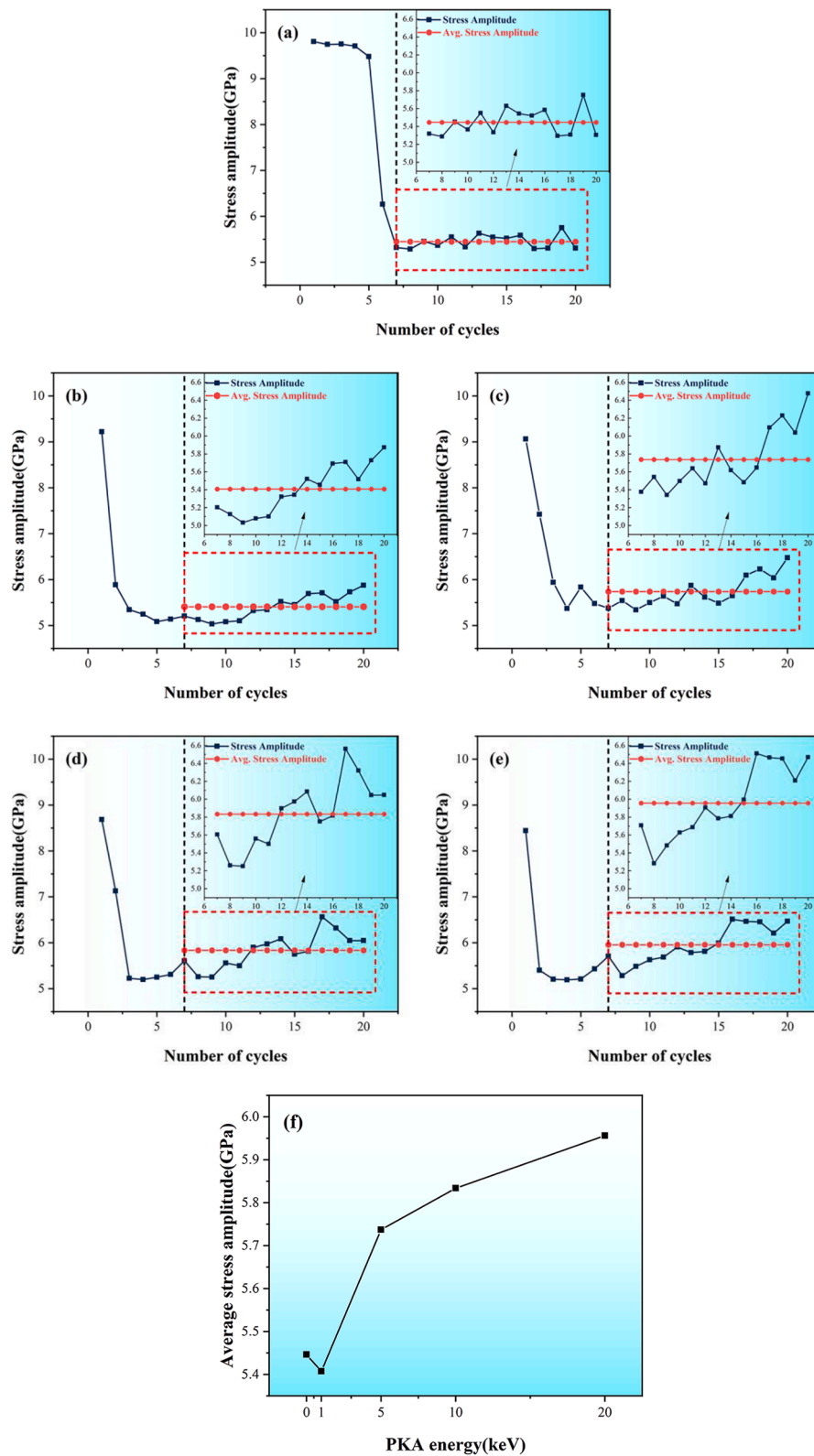


Fig. 8. Variation of stress amplitude with the number of cycles for unirradiated FeCoNiCrCu (a), irradiated FeCoNiCrCu subjected to different PKA energies of 1k eV (b), 5k eV (c), 10k eV (d), and 20k eV (e), and (f) Variation of average stress amplitude with PKA energy.

From the snapshots at key time points recorded in Fig. 10(a), for unirradiated material, no dislocation nucleation was observed in the early stage of cyclic loading (0~25 ps). With the application of reverse compressive loading (30 ps), a small number of dislocations began to appear in the crystal. By 40 ps, dislocations nucleated further and

proliferated, but around 45 ps, most dislocations underwent rearrangement and annihilation induced by cyclic loading. This reversible dislocation evolution behavior persisted in unirradiated samples for approximately five cycles, consistent with its primarily elastic deformation response phase. During this phase, almost no irreversible

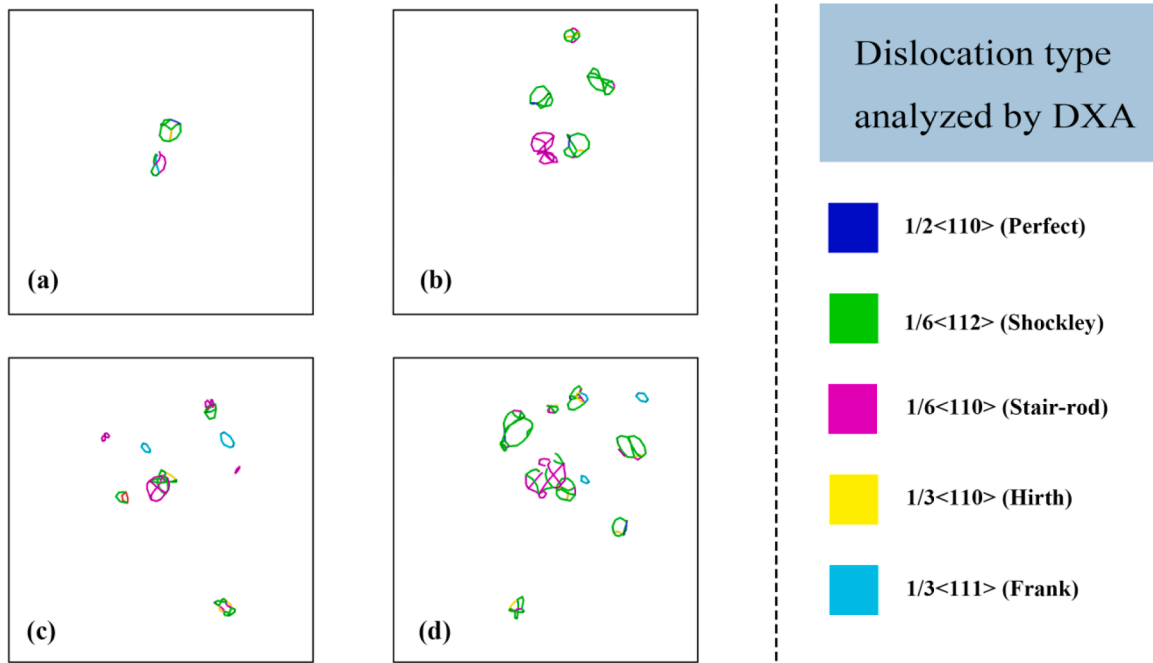


Fig. 9. Dislocation distributions of irradiated FeCoNiCrCu subjected to different PKA energies of 1k eV (a), 5k eV (b), 10k eV (c), 20k eV (d).

damage accumulation occurred within the material, and its mechanical properties remained essentially unchanged after each cycle.

In contrast, irradiated samples already had irradiation-induced dislocation structures at the initial state of cyclic loading (0.0 ps). These defects served as stress concentration points and became the weak zones that preferentially triggered dislocation slip and local plastic deformation, resulting in yielding at relatively lower macroscopic stress and early softening behavior. As shown in Fig. 10(a), at 15 ps and 25 ps, dislocations rapidly nucleated, slipped, and proliferated from defects. With increasing irradiation dose, the initial defect density rose, and dislocation proliferation became more pronounced. Particularly under 20 keV irradiation, dislocation density rapidly increased and spread throughout the crystal during cycling (30.0 ps). Although some dislocations underwent rearrangement and annihilation during unloading and reverse loading between 40–45 ps, a high density of dislocation structures still remained in the samples. Notably, irradiated samples at 45 ps could not achieve complete dislocation recovery like the unirradiated samples, indicating that the dynamic recovery process could no longer restore the material to an entirely elastic state, leading to the accumulation of irreversible plastic strain, macroscopically evident as an earlier yield point in early cycling.

To further reveal the interaction mechanism of irradiated dislocations under cyclic loading, Fig. 10(b) provides an enlarged view of the local defect structure in a 20 keV irradiated sample during the plastic phase. Strong interactions between newly formed dislocations and the pre-existing irradiation dislocation network can be observed, manifesting as entanglement, cutting, and pinning effects among dislocation lines. As deformation progressed, dislocations gradually formed highly localized entangled clusters, effectively hindering further dislocation movement and significantly increasing dislocation slip resistance [54], thereby inducing notable cyclic hardening in subsequent cycles.

In summary, the mechanical response of irradiated FeCoNiCrCu HEA under cyclic loading is a dynamic evolutionary process. It is characterized by an initial softening dominated by irradiation-induced early yielding, followed by significant cyclic hardening triggered by the interaction and entanglement of high-density dislocations.

In face-centered cubic (FCC) materials, plastic deformation primarily originates from dislocation slip, so dislocation density can reflect the

evolution characteristics of the microscale deformation mechanism. The evolution of dislocation density can be statistically analyzed using OVITO, with the dislocation density calculated by the following formula:

$$\rho = L/V \quad (5)$$

Where L is the total length of dislocation lines, and V is the crystal volume. These values can be obtained from OVITO.

The statistical results of dislocation density are shown in Fig. 11. Overall, the dislocation density of all models displayed periodic fluctuations with cyclic loading, but unirradiated and irradiated FeCoNiCrCu samples exhibited distinctly different evolution characteristics.

For unirradiated samples, the change in dislocation density could be divided into three stages. Stage I (the initial five cycles) featured a sharp increase in dislocation density within each loading cycle, followed by near-complete annihilation during unloading, indicating high deformation reversibility. In Stage II, the dislocation density remained at a low level, with minor fluctuations corresponding closely to the stress amplitude fluctuations shown in Fig. 8. Stage III witnessed a continuous rise in dislocation density.

Conversely, the irradiated samples display a two-stage evolution: an initial rapid increase during the first five cycles, followed by a gradual slowdown in the rate of increase, although the overall dislocation density continued to grow. Overall, compared with unirradiated samples, the dislocation density of irradiated FeCoNiCrCu under cyclic loading consistently fluctuated at a higher range. This was mainly due to irradiation damage introducing numerous defects that served as dislocation sources, promoting continuous dislocation proliferation and annihilation during cyclic loading. High density of dislocations enhanced interactions among dislocations, and when they became entangled in stable structures that were difficult to move, this significantly hindered dislocation slip, macroscopically manifesting as cyclic hardening.

3.2.2. Microstructural evolutions

To further explore the plastic deformation mechanisms of FeCoNiCrCu HEA under cyclic loading, this chapter quantitatively analyzed the changes in the fractions of FCC structure atoms and lattice disordered atoms across the cycles. Fig. 12 presents their evolution during the

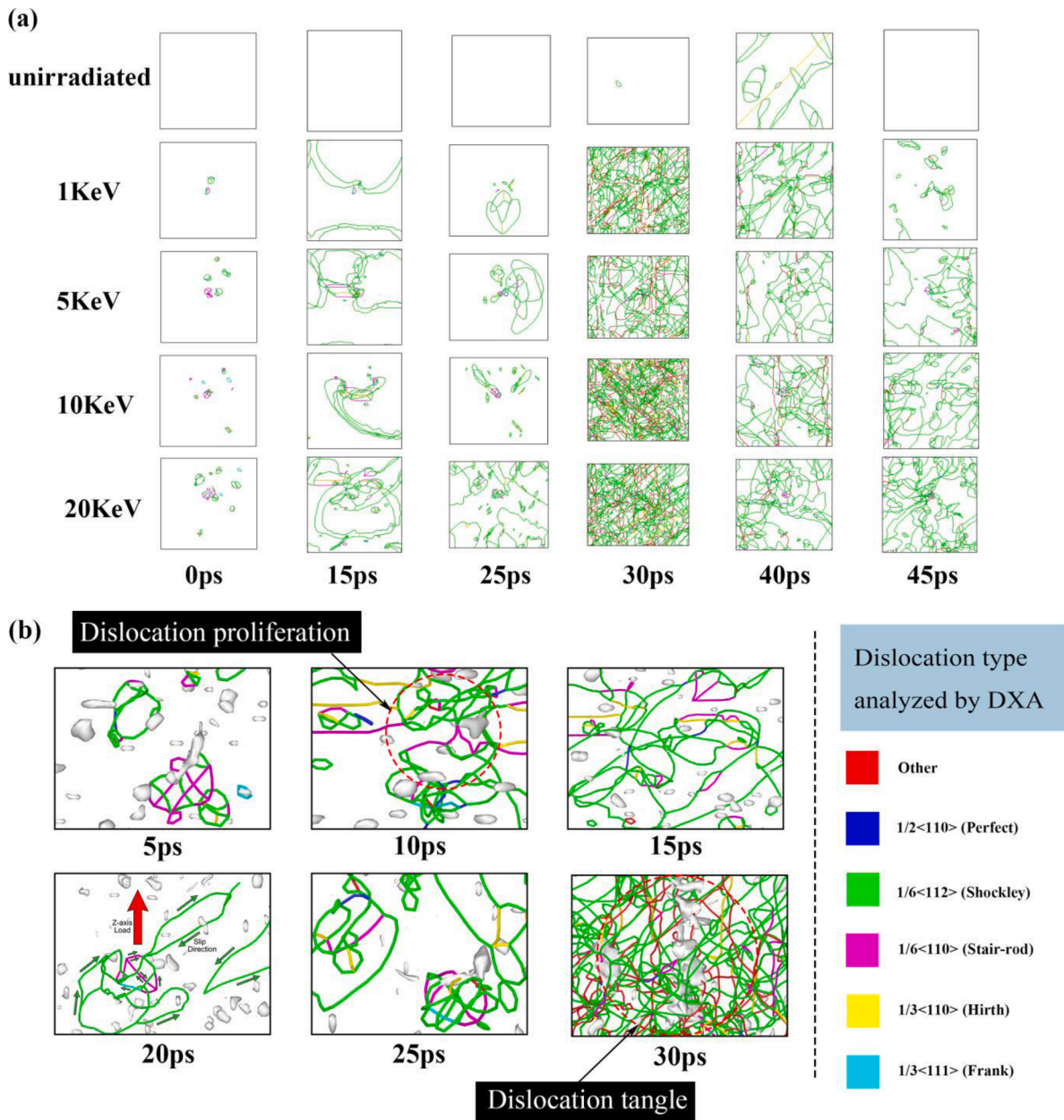


Fig. 10. (a) Dislocation evolutions of different irradiated FeCoNiCrCu at different of the first cycle, (b) Enlarged view of dislocation evolutions of PKA energy 20k eV irradiated FeCoNiCrCu.

cyclic process. As shown in Fig. 12(a), as cyclic loading progressed, the percentage of FCC structure atoms gradually decreased, indicating that some FCC structures transformed into lattice disordered structures or hexagonal close-packed (HCP) phases during deformation. Notably, the higher the irradiation dose, the more significant the decline in the percentage of FCC structure atoms, suggesting that irradiation exacerbated the instability of the crystal structure. Correspondingly, Fig. 12(b) shows that the fraction of lattice disordered atoms continuously increased with the number of cycles, with the 20 keV irradiated sample having the highest disorder atom ratio. The transformation from FCC structures to disordered structures and HCP phases reflected the accumulation of plastic strain during the cycles. In this study, lattice disordered atoms include various crystal defects such as point defects, amorphous atoms, dislocation core atoms, and stacking fault boundary atoms.

It is noteworthy that the trend of lattice disordered atom fraction change was highly similar to the evolution of dislocation density as showed in Fig. 11. Fig. 13 compiles data from five samples. Regardless of

the specific sample conditions, a universal linear relationship is observed between lattice disorder fraction and dislocation density. The global Pearson correlation coefficient of 0.974 confirms that lattice disorder atoms are primary driving force for dislocation evolution in this system. On the one hand, lattice disorder atoms can serve as effective nucleation sites for dislocations during cyclic deformation.

At the same time, these disordered regions can act as obstacles, pinning dislocation motion and inhibiting dislocation recombination and annihilation. In the study by Choi et al. [64], it was found that the slip resistance of dislocations is more significant in lattice disordered regions than in ordered regions. Fig. 14 provides atomic-scale snapshots of the interaction between lattice disordered atoms and dislocations at key moments for the 20 keV sample, illustrating this dynamic process. Before loading (5 ps), irradiation-induced dislocations were pinned and surrounded by clusters of disordered atoms, hindering their motion. Under applied load (10–15 ps), dislocations were forced to bow out or cut through these disordered obstacles, a process that required high

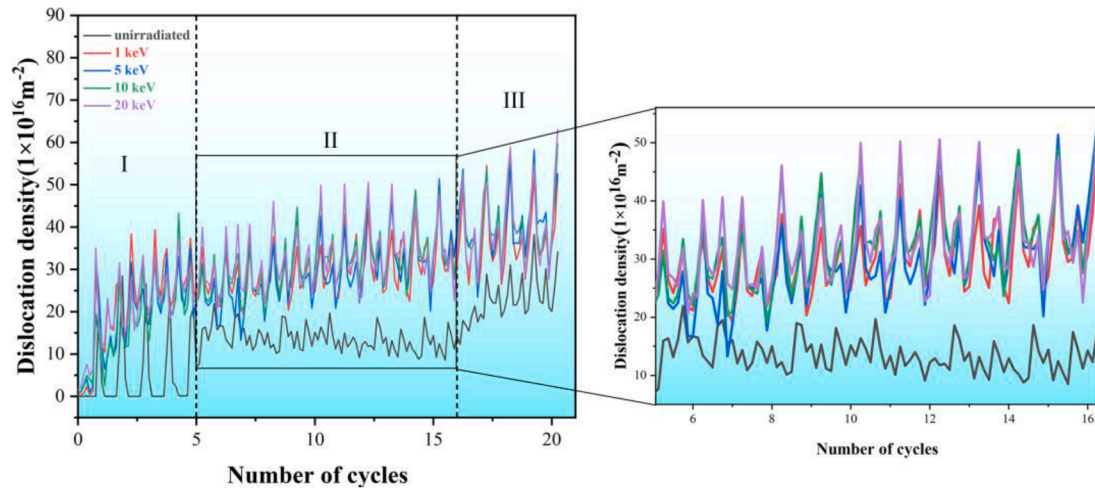


Fig. 11. Variations of dislocation density of different irradiated FeCoNiCrCu with the number of cyclic loading cycles.

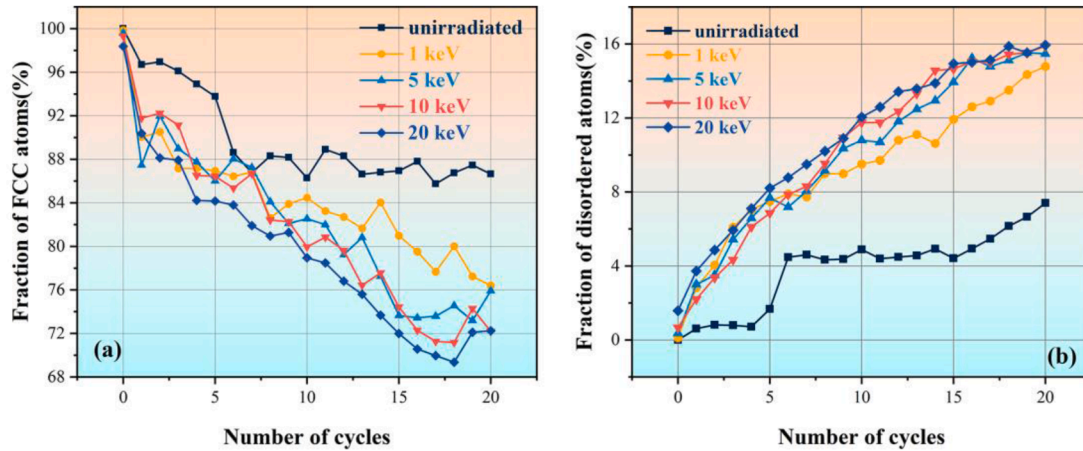


Fig. 12. (a) Variations of FCC atom proportion with the cycle numbers of cyclic loading, (b) Variations of lattice disorder atom proportion with the cycles number of cyclic loading in different irradiated FeCoNiCrCu.

stress and simultaneously destroyed the lattice integrity of the traversed region, generating new disordered atoms. During unloading (20–25 ps), some mobile defects recovered, leading to a temporary reduction in disorder. However, upon reverse loading (30 ps), dislocations from different slip systems or with opposite Burgers vectors encountered each other. Their strong elastic interactions resulted in the formation of stable dislocation tangles. These tangles acted as stronger obstacles, impeding subsequent dislocation motion, and their core regions became stabilized sites with increased local disorder concentration.

To quantitatively verify the pinning effect of disordered domains on dislocation motion, a spatial overlap analysis was performed. An enrichment factor (E-Factor), defined as the ratio of disordered atom concentration within a 5 Å cylindrical zone around the dislocation line (C_{core}) to the global average (C_{avg}), was calculated, as shown in Fig. 15. Throughout the simulation, the E-Factor consistently remains above unity (>1), indicating that the pinning effect of disordered atoms on dislocations is a persistent feature. However, fluctuations in the E-Factor magnitude suggest that while the pinning mechanism remains active, its intensity varies during the cyclic deformation. This fundamentally indicates a strong tendency for disordered atoms to accumulate around dislocation cores, thereby acting as pinning sites that hinder dislocation motion. Ideally, a higher E-Factor correlates with a stronger localized pinning effect, which requires a higher stress to initiate dislocation motion. This is evident in the initial cycles, where the low-dose samples exhibit significantly higher E-Factor values compared to the high-dose

counterparts. This elevated E-Factor indicates a high enrichment of defects at dislocation cores, forming a strong local pinning effect; consequently, a higher applied stress is required for dislocations to overcome this binding and achieve depinning. In contrast, the high-dose samples are characterized by high background disorder, which diminishes the concentration difference between the dislocation cores and the matrix, leading to a lower E-Factor. This effectively lowers the critical barrier required for dislocations to detach from local defect clusters. Furthermore, the temporal evolution of the E-Factor serves as a precise indicator of the material's yielding behavior. The sharp precipitous drop in the E-Factor coincides perfectly with the macroscopic yield point. Physically, this transition corresponds to the onset of dislocation depinning, where dislocations acquire sufficient driving force to overcome the pinning potential imposed by the segregated defect atmospheres. Once the dislocation velocity increases significantly post-yield, the dynamic equilibrium of core segregation is disrupted, causing the localized core concentration (C_{core}) to drop relative to the matrix, leading to the observed collapse in the E-Factor. In the stable cycling stage (cycles 7–20), a distinct bifurcation is observed. All irradiated samples converge to a similarly low E-Factor, suggesting that the continuous interaction with the defect-rich matrix homogenizes the structural contrast. Conversely, the unirradiated sample maintains a relatively higher E-Factor. This is attributed to its pristine matrix background, where any deformation-induced disorder remains highly localized at the dislocation cores, preserving a high signal-to-noise ratio

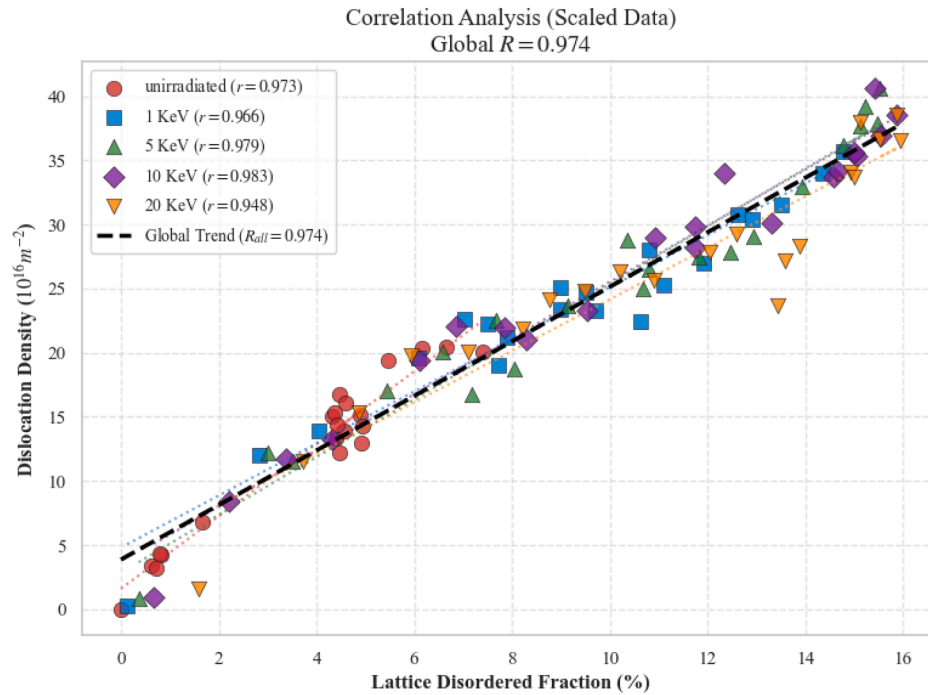


Fig. 13. Correlation between lattice disorder fraction and dislocation density.

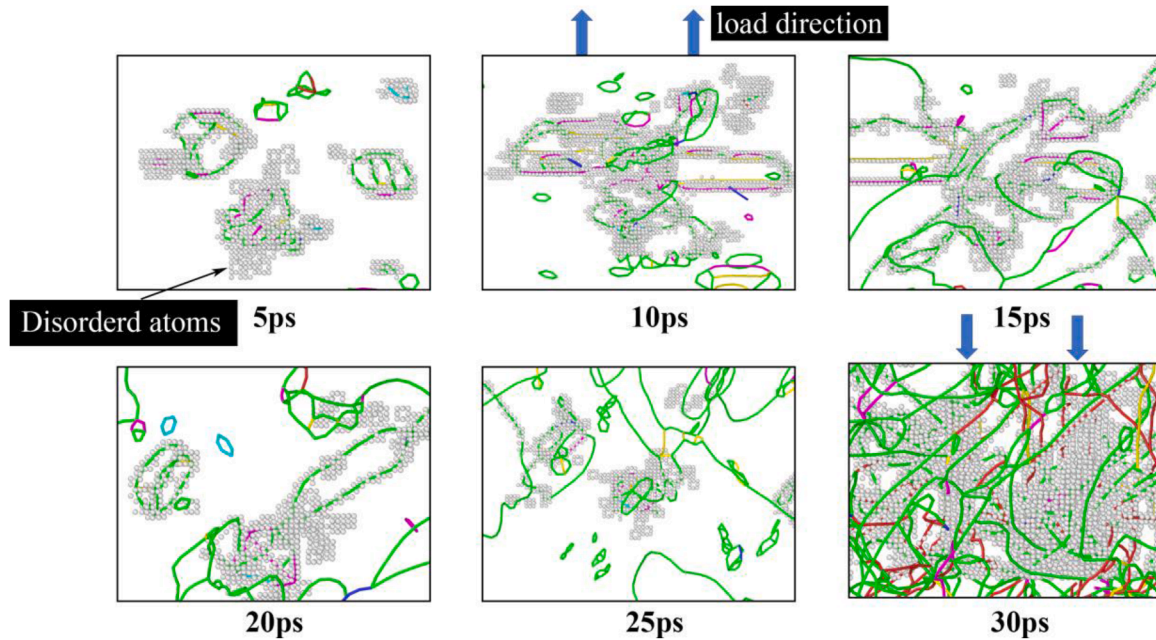


Fig. 14. Interaction between dislocations and lattice disordered atoms under cyclic loading.

in terms of structural heterogeneity.

The findings above indicate that the pinning effect of disordered atoms primarily dominates the strength during the early stage of cyclic loading; however, this mechanism alone cannot fully account for the sustained hardening trend observed in the later stages. To address this and for further analysis of the microscopic evolution during cyclic loading, stacking fault (SF) changes in the FeCoNiCrCu model at different cycle numbers were analyzed, as shown in Fig. 16. In the visualization method used, Common Neighbor Analysis (CNA), atoms are represented in different colors, where green, blue, red, and white atoms represent FCC structures, BCC structures, stacking faults, and

lattice disordered atoms, respectively. The stacking faults are further categorized as follows: the local structure with two layers of HCP atoms is identified as an intrinsic stacking fault, the local structure with one layer of HCP atoms is classified as an extrinsic stacking fault, the local structure with two layers of HCP atoms and two FCC layers is regarded as a deformation twin, and the local structure with three or more layers of HCP atoms is referred to as the HCP-martensite phase [65]. Furthermore, the fraction of HCP atoms (serving a quantitative indicator for stacking faults) was calculated for each snapshot and is annotated below the corresponding model. The comparison of stacking fault defects in different cycles of irradiated models revealed that stacking faults

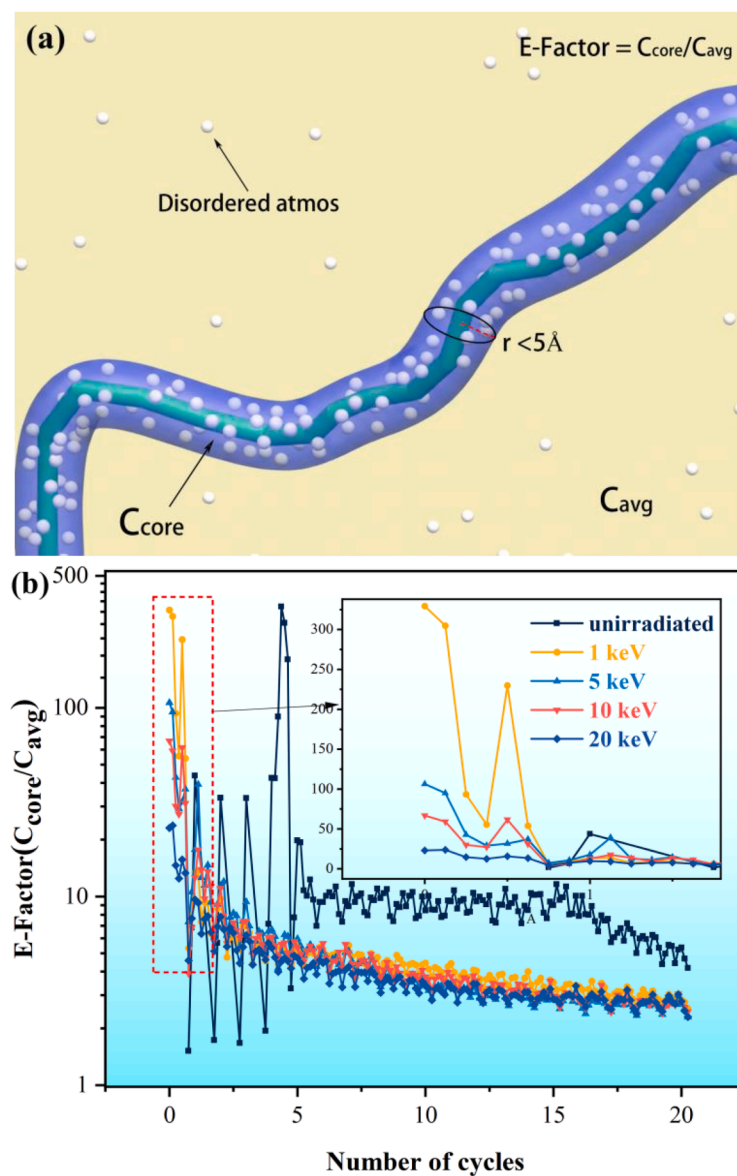


Fig. 15. Definition and evolution of the enrichment factor (E-Factor). (a) Schematic representation of the calculation model, showing disordered atoms within the cylindrical shell surrounding the dislocation core. (b) Variation of the E-Factor during the loading process.

became more significant with increasing cyclic loading. This indicated that stacking faults are an important form of defects under cyclic loads.

Additionally, it is important to note that the number of defects in unirradiated models was significantly less than in irradiated models, and the number of stacking faults did not consistently grow, showing fluctuating changes, especially a noticeable decrease at $N = 20$. This fluctuation in microstructure directly corresponded to the macro mechanical performance response: the stress amplitude of unirradiated FeCoNiCrCu exhibited a fluctuating change. In contrast, the continuous accumulation and retention of stacking faults in irradiated samples corresponded to an overall stable increase in stress amplitude.

Table 3 provides a quantitative assessment of the microstructural and mechanical evolution from Cycle 10 to Cycle 17, capturing the critical divergence point in the cyclic response. At Cycle 10, the samples exhibit comparable varying initial states, but the subsequent evolution reveals fundamentally different deformation mechanisms. A striking observation is the saturation behavior of the unirradiated sample. Between Cycle 10 and Cycle 17, its HCP phase fraction remains completely stagnant at 9.5 %, while the stress amplitude actually experiences a slight decrease from 5.36 GPa to 5.29 GPa, indicating a lack of further

hardening drive. In sharp contrast, all irradiated samples exhibit a vigorous activation of phase transformation and hardening during this interval. The high-energy irradiation samples, in particular, display the most aggressive evolution. For instance, the 20 keV sample sees its HCP fraction surge from 10.0 % to 12.9 %, accompanied by a significant stress increase from 5.63 GPa to 6.46 GPa. Similarly, the 10 keV sample reaches a peak stress of 6.56 GPa at Cycle 17, jumping from 5.56 GPa at Cycle 10. It is worth noting that while the instantaneous stress at Cycle 17 is marginally higher for the 10 keV sample, the 20 keV sample maintains a higher overall average stress amplitude throughout the cyclic process, consistent with the higher defect density induced by higher irradiation energy. This comparison conclusively demonstrates that while the unirradiated material reaches a stable state early on, the irradiation-induced defects in the treated samples continue to drive massive stacking fault generation (f_{HCP} increase) well into Cycle 17, thereby sustaining a rapid and continuous cyclic hardening process.

This macroscopic disparity stems from distinct deformation mechanisms. In unirradiated materials, the absence of stable pinning centers permits dynamic recovery: dislocations undergo cross-slip and annihilation, causing the periodic disintegration of SF networks and resulting

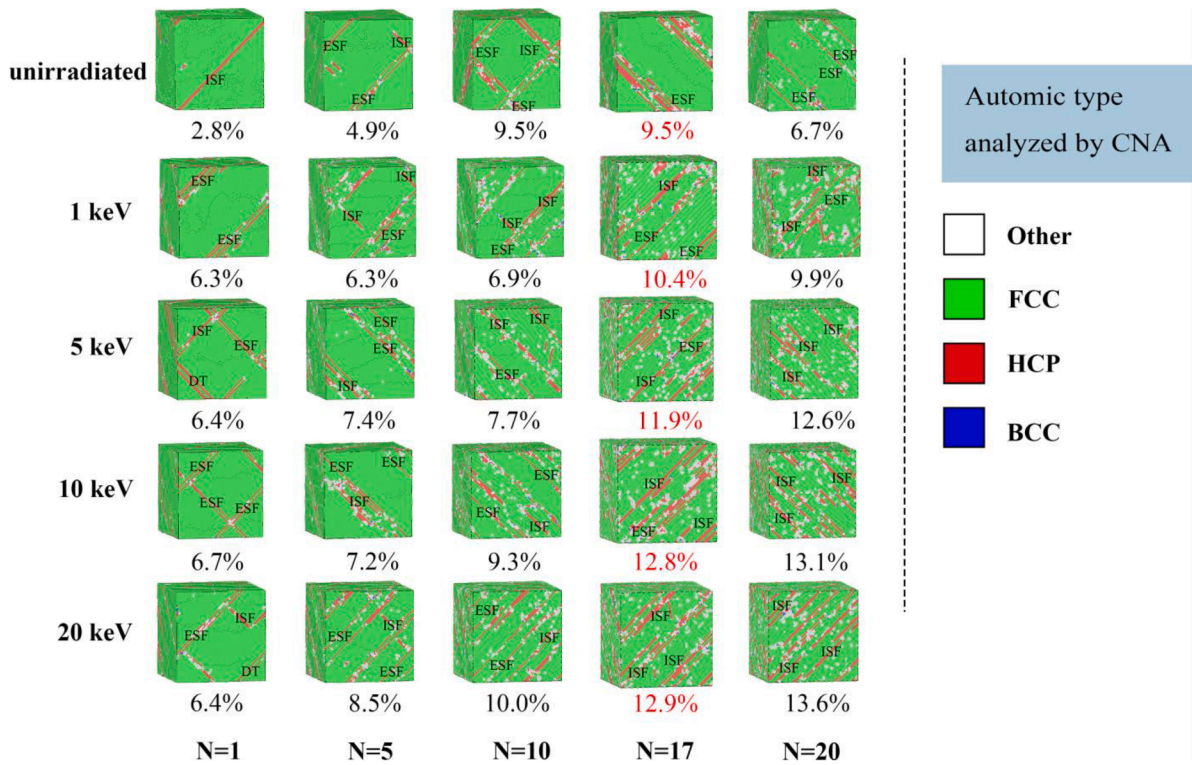


Fig. 16. Stacking fault defect distributions of different irradiated FeCoNiCrCu at different cycles.

Table 3

Quantitative comparison of HCP phase fraction (f_{HCP}) and stress amplitude (σ_{amp}) between the intermediate stage (Cycle 10) and the critical transition stage (Cycle 17).

Sample	Cycles 10 f_{HCP} (%)	σ_{amp} (GPa)	Cycle 17 f_{HCP} (%)	σ_{amp} (GPa)
unirradiated	9.5	5.36	9.5	5.29
1 keV	6.9	5.08	10.4	5.71
5 keV	7.7	5.50	11.9	6.09
10 keV	9.3	5.56	12.8	6.56
20 keV	10.0	5.63	12.9	6.46

in fluctuating stress amplitudes. In irradiated samples, however, stable irradiation defects inhibit such recovery. As noted by Li et al. [66], the continuous accumulation of planar defects (SFs) creates extensive barriers that effectively impede dislocation slip. The increasing SF density necessitates a higher driving force for deformation, thereby sustaining the rapid and continuous cyclic hardening process.

3.3. Limitation and outlook

It should be noted that cyclic deformation additionally involves longer-timescale processes, including dislocation reciprocity, annihilation, and progressive cell-structure formation. In the present simulations, however, the cyclic simulations were performed at a high strain rate of $1 \times 10^{10} \text{ s}^{-1}$. Since dislocation motion is inherently time-dependent, such a rate reduces the time available for thermally activated recovery during load reversal, which may suppress dislocation annihilation and limit dislocation reversibility, thereby promoting the retention of defects generated in each cycle. In addition, the short timescale may hinder the development of well-equilibrated dislocation cell structures that typically require longer-range dislocation rearrangements. Meanwhile, the high rate may promote the formation and retention of stacking faults and related sessile junctions, thereby enhancing the apparent hardening response. Therefore, although

quantitative metrics such as absolute stress amplitudes, overall dislocation densities, and the rate of SF accumulation inevitably carry rate-dependent biases and are likely overestimated, the qualitatively identified mechanism of SF-network-assisted strengthening mediated by partial-dislocation activity remains credible from a mechanistic perspective.

Overall, these findings provide mechanistic insight into the cyclic plasticity of irradiated FeCoNiCrCu. Distinct from previous MD studies that mainly examined the monotonic tensile behavior of irradiated HEAs, where irradiation was primarily manifested as earlier yielding [48], the present work establishes the dose dependence of the cyclic response and elucidates the underlying defect-dislocation interactions. Furthermore, it reveals a transition in the dominant hardening mechanism, from atomic-disorder pinning at early cycles to stacking-fault-controlled hardening at later cycles. In contrast to the cyclic softening associated with dislocation channeling reported for irradiated RAFM steel EUROFER97 [17], irradiated FeCoNiCrCu exhibits sustained cyclic hardening, indicating a fundamentally different irradiation-fatigue interplay in this HEA system. Future work combining lower effective loading rates, longer cycle durations, and larger simulation cells, as well as systematically examining the effects of different strain rates, would be valuable for further assessing the long-timescale structural evolution under cyclic deformation.

4. Conclusion

This study employed molecular dynamics simulations to investigate the atomic-scale deformation mechanisms and microstructural evolution of FeCoNiCrCu HEAs under varying irradiation doses, aimed to understand the fatigue performance of HEAs under high-dose irradiation. Overlapping cascade simulations were conducted at PKA energies of 0, 1, 5, 10, and 20 keV to construct defect models. Uniaxial tensile tests were applied to the defect models to examine their mechanical properties, followed by cyclic loading to simulate the material's low-cycle fatigue. The simulation results lead to the following main

conclusions:

- (1) The stress amplitude of the FeCoNiCrCu alloy exhibits a positive correlation with irradiation dose, a behavior primarily governed by the evolution of dislocation density. Irradiation-induced defects act as preferential dislocation nucleation sites, leading to earlier activation and a higher dislocation population than in the unirradiated alloy; the resulting dislocation interactions impede glide and produce pronounced hardening.
- (2) During the initial stage of deformation, disordered atoms play a dual role that governs early-stage strength: they act as nucleation sites for dislocations and, more critically, serve as strong pinning centers that hinder dislocation slip. Consequently, dislocations undergo intermittent pinning–depinning during glide, elevating the early-stage flow stress.
- (3) As deformation progresses, the dominant hardening mechanism transitions to stacking faults (SFs). The metastable nature and low SFE of the FeCoNiCrCu matrix facilitate the formation of a high density of SFs. These planar defects constitute extensive barriers within the distorted lattice, impeding subsequent dislocation motion more effectively than point defects and serving as the primary source of sustained cyclic hardening in the later stages.

The observed cyclic hardening implies a critical reduction in plastic accommodation capability. This behavior is driven by the pinning effect of disordered atoms and extensive stacking fault networks. Due to the alloy's low stacking fault energy, the dissociation of dislocations into Shockley partials suppresses cross-slip, preventing stress relaxation via dynamic recovery. Consequently, restricted dislocation mobility creates persistent atomic-scale stress concentrations. Instead of dissipating through plastic flow, these unrelaxed local stresses serve as preferential sites for bond breaking or void nucleation, thereby directly facilitating crack initiation and early-stage failure.

Due to the high strain rates ($1 \times 10^{10} \text{ s}^{-1}$) adopted in the present MD simulations, the reported absolute stress amplitudes and the quantitative accumulation of dislocations and stacking faults may carry an upward bias. Nevertheless, the qualitative identification of the transition in cyclic deformation mechanisms from defect pinning to stacking-fault-controlled hardening remains qualitatively valid.

CRediT authorship contribution statement

Zhanguang Zheng: Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Funding acquisition, Data curation, Conceptualization. **Yunjia Mo:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Conceptualization. **Liping Jiang:** Validation, Software, Investigation, Formal analysis. **Xiangyu Ruan:** Visualization, Methodology, Investigation. **Yangguang Hao:** Software, Methodology, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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